Vertex Operators and Symmetric Functions

Introduction to Modern Advances in Algebra

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Vertex Operator

Heisenberg algebra: $\mathfrak{h} = \operatorname{span}\{a_n\}_{n\in\mathbb{Z}}$

$$[a_n, a_m] = n\delta_{n+m,0}$$

a_n are the modes of the affine vertex algebra associated to $\mathfrak{gl}_1$

Part of the data of this vertex algebra is its vertex operator:

$$V(z) = \exp\left(\sum_{n>0} \frac{1}{n} z^n a_{-n}\right) \exp\left(\sum_{n>0} \frac{1}{n} z^{-n} a_n\right)$$

Symmetric Functions

Λ: Ring of symmetric functions in $x_1, x_2, \ldots$

$p_n = x_1^n + x_2^n + \cdots$ (power sums)
$\Lambda \cong \mathbb{C}[p_1, p_2, \ldots]$

$\mathfrak{h} \curvearrowright \Lambda$: for $n \in \mathbb{Z}_{>0}$

$$a_{-n} f = p_n f, \qquad \text{and} \qquad a_n f = n \frac{\partial}{\partial p_n} f.$$

Fermionic Fock Space

Maya diagram: row of black and white dots (“go stones”) with appropriate boundary conditions, eg

$$\cdots\ \bullet\ \bullet\ \bullet\ \bullet\ \circ\ \bullet\ \circ\ \bullet\ \circ\ \circ\ \bullet\ \circ\ \circ\ \circ\ \cdots$$

$\longleftarrow$ all black   all white $\longrightarrow$

$\mathcal{F}$ is the span of all such diagrams

$\mathfrak{h} \curvearrowright \mathcal{F}$, eg, acting on diagram above with a_{-2} gives

$$-\ \cdots\ \bullet\ \bullet\ \circ\ \bullet\ \bullet\ \bullet\ \circ\ \bullet\ \circ\ \circ\ \bullet\ \circ\ \circ\ \circ\ \cdots$$
$$+\ \cdots\ \bullet\ \bullet\ \circ\ \bullet\ \circ\ \bullet\ \circ\ \circ\ \circ\ \bullet\ \bullet\ \circ\ \circ\ \circ\ \cdots$$
$$+\ \cdots\ \bullet\ \bullet\ \bullet\ \bullet\ \circ\ \bullet\ \circ\ \bullet\ \circ\ \circ\ \circ\ \circ\ \bullet\ \circ\ \cdots$$

Theorem (Boson-Fermion Correspondence)

The $\mathfrak{h}$-modules Λ and $\mathcal{F}$ are isomorphic

Vertex Operators

$h_n = \sum_{i_1 \leq \cdots \leq i_n} x_{i_1} \cdots x_{i_n}$ (complete homogeneous)

$h_1 = x_1 + x_2 + \cdots$, $h_2 = x_1^2 + x_2^2 + x_1 x_2 + \cdots$

Generating function: $H(z) = \sum_{n \geq 0} z^n h_n = \prod_{i \geq 1} \frac{1}{1 - z x_i}$

By identities between symmetric functions

$$H(z) = \exp\left(\sum_{n>0} \frac{1}{n} z^n p_n\right)$$

which corresponds to the action of the $\mathfrak{h}$ half vertex operator

$$\exp\left(\sum_{n>0} \frac{1}{n} z^n a_{-n}\right)$$

References

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[Mac98] Ian MacDonald. *Symmetric Functions and Hall Polynomials*. 2nd ed. Oxford University Press, 1998.